



FINTECH LENDERS IN THE MORTGAGE MARKET

*As an assessment for the Banks & the Fintech Revolution
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1. Introduction

The nature of mortgage lending has evolved since the global financial crisis. In the Pre-crisis era, development was characterized by proprietary technology within large, regulated financial institutions. However, the 2008 crisis eroded public trust in traditional financial Institutions. Post-GFC pressures, including profitability constraints and regulatory reforms in the U.S (e.g., Dodd-Frank Act) which curtailed innovation among incumbents while elevating costs and reducing accessibility of financial services. Hakida, et al (2019) Note that this lending environment catalysed a decisive pivot toward decentralization finance and hence the rapid emergence of FinTech startups. Against this backdrop, fintech credit platforms (peer-to-peer lending, invoice trading) and big tech firms emerged as significant credit intermediaries.

Studies (Giglio, 2021; Alois, 2015) show that the levels of public trust in tech companies managing personal finance is not only increasing, but outstripping the trust in traditional lenders. For example, the level of trust Americans have in CitiBank is 37%, while the trust in Amazon and Google is seventy-one percent and sixty-four percent, respectively. In addition to these tech-goliaths, findings (Weihuan Zhou, Douglas W. Arner, and Ross P. Buckley, 2015) show that the bulk of grassroot adoption of alternative lending services is met by young start-ups with over 2000 P2P lending platforms operating outside a clear default framework.

Beyond eroding public trust, Neobanks and Fintechs address a long-standing gap in the American financial landscape. Abbas et al. (2025) reveals that 4.5% of urban households in the U.S remain unbanked, as well as an additional 20% of Americans who depend on alternative financial services for basic transactions. New migrants, young adults and minority groups making up the bulk of this underbanked statistic. (Cosma & Rimo, 2023; Risbabh, 2024) believe that while traditional banks, constrained by rigid financial criteria in their operational models (i.e reliance on credit bureau data), struggle to adapt their services for financially marginalized populations. FinTechs, have transformed financial services by leveraging technology to make finance more affordable and accessible; particularly for first-time homebuyers and underserved demographics. Essentially, they scratch an itch that traditional banks cannot reach. This disruptive potential has not gone unnoticed: major banks are increasingly venturing into the FinTech space through acquisitions and investments. For instance, since 2023, JPMorgan has allocated \$18 billion specifically for FinTech investments, acquiring companies like: Global Shares for \$730 million, A stake in Viva Wallet (A European payments giant) for \$800 million, Aumni (investment analytics software), and Frank, a startup which targeted students and young people.

The Earliest Fintechs in the U.S were built to circumvent troublesome payment solutions. An example of which is PayPal, which launched in 2000 and surged into prominence on the backbone of online payments. However, fintechs did not veer into the business of lending until Prosper Marketplace and Lending Club were established in the middle 2000s. Prosper launched in 2006 as the first p2p lending platform in the U.S. Initially, they operated with a very different model which allowed borrowers with subprime credits scores access to facilities with interests rates better than what mortgage banks offered. Kim (2009) recounts that this first iteration became known as Prosper 1.0 and it was not until 2009 when they reopened after their quiet period that Prosper became a lending platform focused on prime and near prime borrowers. In its early years, Prosper operated a variable rate model, online auction marketplace, with lenders and borrowers ultimately determining loan rates using a Dutch auction-like system. Quicken Loans (now rebranded as

Rocket Mortgage) emerged the largest single fintech mortgage lender in the U.S. According to a 2021 HMDA (Home Mortgage Disclosure Act) data, Quicken Loans originated more than 1.1 million loans in 2020.

Regardless of their strides across the various financial sub-industries, scholars (Buchak et al. 2018a; Fuster et al. 2019) seems to agree that the mortgage lending space is a new frontier for fintech ventures as traditional banks and credit unions remain the first-choice service providers in the category. It remains unclear whether these technology-equipped tech lenders can fill the credit gap or make a lasting impact in (disrupt) the industry.

This essay examines the expanding role of fintech lenders in the U.S. mortgage market. Specifically, we analyze and compare lender distribution, borrower credit profiles, interest rate structures, delinquency performance, and predictive modeling of loan outcomes, while comparing findings to existing literature and recent empirical studies.

1.2 Literature Review

This section reviews previous scholarly articles and literature. Research in this area explores FinTech's role in credit access, risk assessment, operational efficiency, and financial inclusion, often comparing it with traditional lending models.

(Jagtiani & Lemieux, 2019; Cosma & Rimo, 2021; Abbas, Abid & Rafique, 2025) compared fintechs to traditional bank and non-bank entities, they found that personal loans originated through fintech platforms could generate significant savings to borrowers when compared with similar traditional loans, in part because of lower interest rates driven by a more accurate representation of credit risks.

Cornelli, et al (2020), opines that fintech lending is more developed in areas where there are fewer bank branches per capita (i.e. rural areas). Adding that loans charged by fintechs in the U.S typically ranged from 9 - 28%, they insist that fintech as a reward for their superior innovative technology would have lower rates of delinquency across developed countries. Likewise Hakida & Perry (2019) agree with this postulation, arguing that the fintech's larger bouquet of financing options would reduce delinquency.

Fuster, A., Plosser, M., Schnabl, P., & Vickery, J. (2019). show that mortgage credit could be more efficiently priced using machine learning, but they highlight concerns about such methods generating statistical discrimination, they also found that fintech lenders like non-fintechs tend to apply higher interest rates to borrowers from minority backgrounds. These finding align with Buchak, G., Matvos, G., Piskorski, T., & Seru, A. (2018) additionally found that regulatory burdens and technological improvements have contributed to the decline of traditional banks' market share.

These studies all agreed that fintech's advanced innovative technology would increase access to finance to sub-prime and near-prime borrowers while having lower cases of delinquent loans.

2. Methodology & Data Summary

This section discusses the dataset employed in this study, and analyses trends in fintech and Non-fintech credit markets. In particular, it describes the Freddie Mac data source used and necessary choices on data aggregation, cleaning and estimation.

The Federal Home Mortgage Corporation was established in 1970 by the U.S congress to operate as a Government Sponsored Enterprise. The Freddie Mac Single Family Loan-Level Dataset includes information on the origination and monthly loan-level performance of millions of fixed-rate, fully amortizing mortgages purchased or guaranteed by Freddie Mac. The default origination dataset captures 31 key variables on borrower attributes, loan terms, property details and lender types.

2.1 Summary Statistics and Data Preparation

We employed data preparation and cleaning processes to ensure the reliability of the analysis. The Freddie Mac's loan origination and servicing records, initially contained over one million observations across 61 variable-columns.

First, missing values were addressed by removing columns with excessive null entries (e.g., `flag_fthb`, `flag_sc`), as these variables lacked sufficient data for meaningful interpretation. Additionally, placeholder values such as 9999 (FICO scores), 999 (DTI, CLTV, LTV), and 9 (property valuation codes) were identified and filtered out, as they represented missing or invalid entries rather than true numerical values. This step was crucial to prevent skewed statistical summaries and erroneous model training.

After merging origination and servicing data, further cleaning involved dropping rows with null values to maintain consistency. A delinquency flag variable (`delinquency`) was then created to classify loans based on their repayment status, this allowed us to have a category of loans for just delinquent loans (1 for delinquent loans and 0 for non-delinquent loans), facilitating subsequent risk analysis.

Finally, feature engineering steps—such as log-transforming skewed variables (e.g., DTI, interest rates) and encoding categorical lender types—were applied to improve model performance. These preprocessing measures ensured the dataset's integrity, enabling robust comparisons between FinTech and traditional lending outcomes.

The unavailability of first-time-home-owner flag, income, gender, immigration status et cetera means that we could not analyze if fintechs truly have a significant share of young home buyers and underserved communities.

Description Statistics and Visualizations

	fico	dt_first_pi	dt_matr	mi_pct	cnt_units	cltv	dti	orig_upb	ltv	int_rt
count	28167.000000	28167.000000	28167.000000	28167.000000	28167.000000	28167.000000	28167.000000	28167.000000	28167.000000	28167.000000
mean	746.378209	201721.870629	204339.711151	5.751092	1.031810	73.013597	34.818937	220940.462243	72.706003	4.145449
std	46.910609	33.957008	636.849157	11.036579	0.241264	16.848808	9.529879	117631.209765	16.899919	0.462725
min	549.000000	201702.000000	202504.000000	0.000000	1.000000	6.000000	1.000000	17000.000000	6.000000	2.250000
25%	713.000000	201705.000000	203711.000000	0.000000	1.000000	65.000000	28.000000	130000.000000	64.000000	3.875000
50%	754.000000	201708.000000	204705.000000	0.000000	1.000000	78.000000	36.000000	200000.000000	77.000000	4.125000
75%	786.000000	201711.000000	204709.000000	0.000000	1.000000	80.000000	43.000000	294000.000000	80.000000	4.375000
max	832.000000	201810.000000	204809.000000	35.000000	4.000000	97.000000	50.000000	967000.000000	97.000000	6.125000

Table 2.1: Descriptive Statistics of numerical values (post cleaning)

Table 2.1 provides a descriptive excerpt (count, mean, standard deviation, min, 25th percentile, median, 75th percentile, max) across 14 numerical variables and 28,167 observations. The fico score section suggests that the participants are typically within the prime-lending category according to the characterization of Toh Y. L. (2016). A further analysis of this would be discussed later on. Debt to income (dti) mean of 34.82 suggests that most borrowers exceed the conventional threshold of 33.33% (or a-third of their income). Interest rates show a Mean of 4.15%, Ranging 2.25% → 6.13% This reflects typical pre-covid and first Trump-era rates. Loan sizes (orig_upb) varies widely, with the concentration of which being in the 294k to 130k value, it ranges from as little as \$17,000 to as much as \$967,000 with a mean of \$220,940.

Bar-chart of Dataset

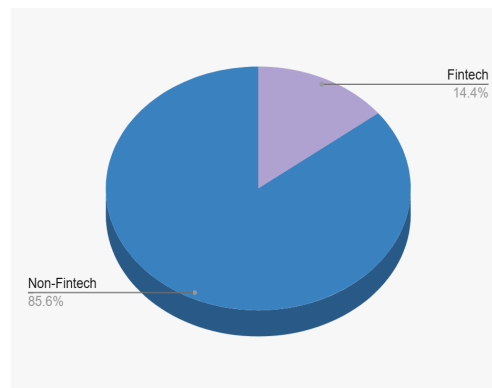
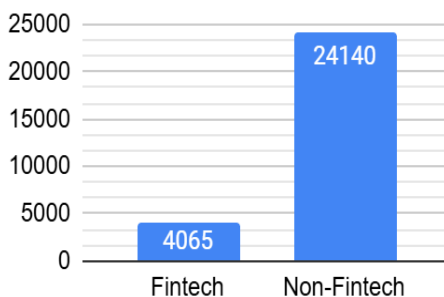


Figure 2.1: Distribution of Fintech V Non-Fintech

Our second graph shows that fintech lenders account for approximately 14.4% of the dataset, while non-fintech (traditional) lenders dominate at 85.6%. These figures corresponds to the finding from a 2017 Bank of International Settlement working paper estimated loans by fintech and bigtech lending to account for 14% of all loans disbursed in the year 2019 in the U.S. Other studies (BIS, 2020; Jagtiani & Lemieux, 2019) hints at slightly higher trends in non-mortgage facilitations, noting fintechs' rapid expansion due to digital innovations and consumer preferences for faster, more convenient loan processing for other loan types. This suggests that while fintech adoption is increasing, the mortgage market might experience this change afterwards.

2.2.2 Credit Score (FICO) Distribution

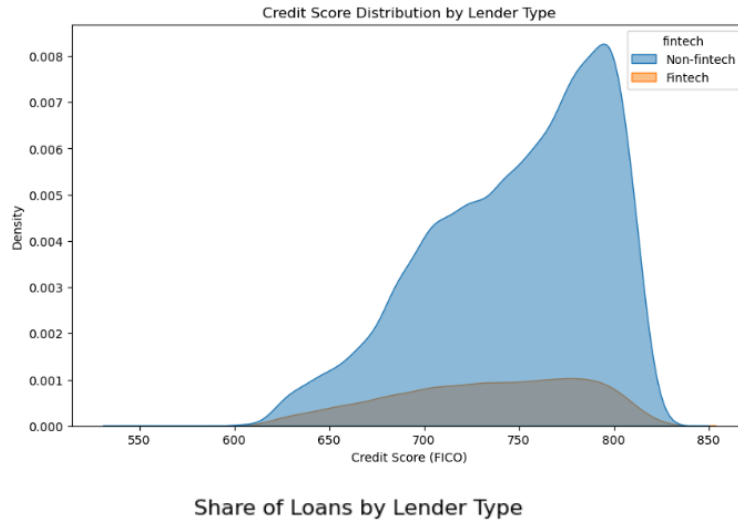


Figure 2.2 Density plot; Share of Loans by Lender Type

Figure 2.2 illustrates the distribution of borrower credit scores (FICO) segmented by lender type (FinTech vs. non-Fintech), confirming a wider spread of credit scores among non-fintech borrowers with the highest concentration closer to 800 score. Fintech lenders have a flatter curve meaning that they serve slightly riskier borrowers on average:

Our analysis further show that on average the credit scores are;

Risk Category	Near Prime	Prime	Sub-prime	Super-prime	Total
Lender Type					
Fintech	364(8.9%)	2849(70.2%)	0	846(20.8%)	4059
Non-Fintech	1128(4.7%)	15452(64.1%)	2(0.1%)	7526(31.2)	24108

Table 2.2: Distribution of Loans by Credit score.

Fintech lenders: Average FICO score = 733.72, prime lending category Non-Fintech lenders: Average FICO score = 748.51 This aligns with FinTechs' disruptive role in expanding credit access to underserved segments, albeit at slightly higher risk.

Utilizing the credit scoring system employed by Toh Y. L (2023) We categorize loans as; Super-prime (>780), Prime (661-780), Near-prime (601-660), Subprime (≠600) to obtain a granular understanding of the distribution.

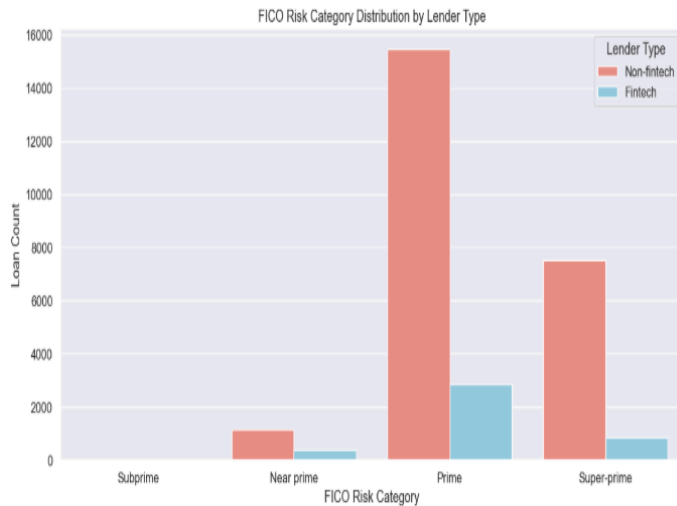


Figure 2.3 A Bar-chart Loan by Fico Score

Table 2.2 further illustrates our findings from figure 2.2 showing that compared to non-fintechs, traditional banks tend to have a preference for higher credits or super-primes as these groups make up 31% of their loan ratio in the dataset compared to about 21% among non-fintechs, suggesting that non-fintechs tend to have a preference for safer and higher quality bets, existing literature (ADD EXISTING LITERATURE HERE) suggests that this could be because fintechs have a higher reliance on existing credit grading system which puts excluded groups at a disadvantage. Adversely, approximately 70% of fintech loans in the dataset fall into the prime category, compared to 64% for non-fintechs, suggesting that fintechs cater more aggressively to underserved segments of borrowers. Fintechs however do have a considerably higher percentage of loans (8.9% compared to 4.7%) among near-prime borrowing, this agrees with existing literature (Abbas et al; 2025, Giglio F; 2021, Jagtiani, J., & Lemieux, C.; 2018, Fuster et al; 2019), who observed fintechs' higher tolerance for slightly riskier profiles, possibly due to superior credit scoring models using alternative data, allows them to service groups underserved by credit unions and mortgage banks. Our graph also shows that both fintech and non-fintech lenders tend to avoid individuals with subprime credit scores.

2.2.3 Interest Rate Comparison

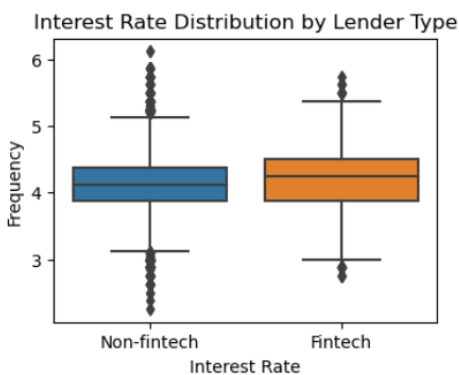


Figure 2.4: Average Interest rate Distribution

- Fintech lenders: Average Interest Rate = 4.20%
- Non-Fintech lenders: Average Interest Rate = 4.14%

Average interest rates tend to be generally higher for fintech-originated loans. Considering that credit scores generally have a directly inverse impact on interest rates, (i.e higher credit-scores highly safely and attract lower interest rates) for fintechs this could potentially mean higher returns on assets as they are able to charge slightly more while keeping overheads and costs lower. It is interesting to note that the average interest rate for Non-fintech loans is closer to the general average of 4.15% from Table 2.1. The difference of 0.06 complies with the findings of Buchak, et al (2018), who found negligible differences between Fintech and Non-fintech interest rates.

2.2.4 Distribution By Service Provider

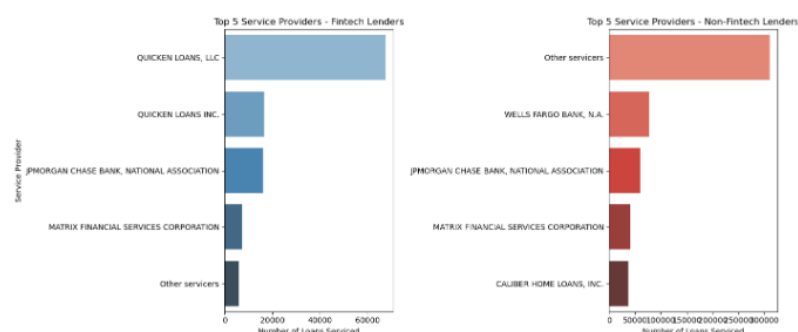


Figure 2.5 shows the analysis of loan servicing reveals distinct operational models between FinTech and non-FinTech lenders. FinTech loans are overwhelmingly serviced by Quicken Loan, which suggests a vertically integrated or highly coupled lending-servicing model. This integration can lead to streamlined operations, consistent customer experience, and better control over the entire loan lifecycle. In stark contrast, non-FinTech lenders exhibit a more fragmented servicing structure, with a substantial proportion of their loans being handled by a diverse group of "Other servicers". This fragmentation suggests that traditional lenders may rely more heavily on third-party or regional servicing platforms. These differing servicing strategies carry significant implications for operational efficiency, the borrower experience, and overall risk management. A more integrated model, as seen with FinTechs, might offer enhanced data flow and quicker response times, while a fragmented model might provide greater flexibility in scaling and specialized regional expertise.

It's also worth noting that entities like The Matrix Financial Services Corporation (MSR Holdings LLC) actively purchase the rights to service mortgage loans on behalf of Fannie Mae/Freddie Mac, highlighting the secondary market for loan servicing. Likewise the appearance of JP Morgan Chase Bank as a prominent fintech lender can be as a result of their underwriting of loans from acquired fintech entities.

2.2.5 Distribution By Loan Amount.

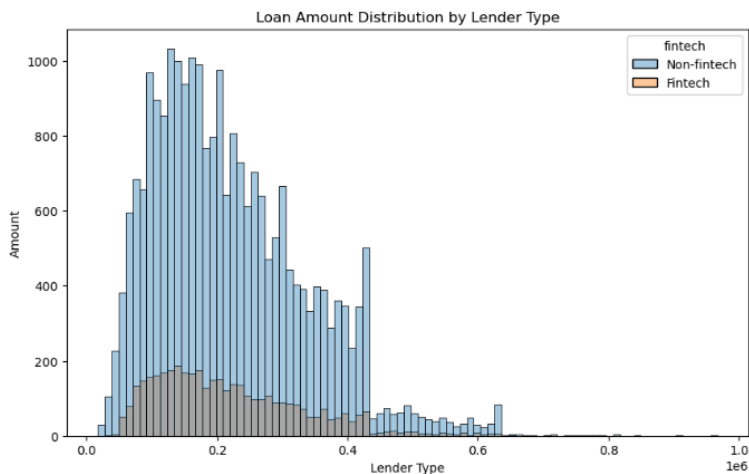


Figure 2.5 Histogram: Distribution by Loan Amount (Fintech V Non-fintech)

Figure 2.5 illustrates the loan amounts, revealing a natural distribution with the both extremities tapering off and clear distinctions between FinTech and non-FinTech lenders with regards to the loan size. Non-FinTech lenders demonstrably dominate the market in terms of both loan volume and the diversity of loan sizes, particularly in the higher-value segments. This suggests their capacity to underwrite and manage a broader spectrum of larger mortgages. Conversely, FinTech lenders primarily issue smaller loans, with a pronounced concentration in the \$1,000 to \$200,000 range. This pattern suggests that FinTechs may be targeting a specific borrower demographic; perhaps those seeking faster approvals for smaller loan amounts, or they might face structural limitations in underwriting significantly larger loans due to capital constraints, risk appetite or more likely due to the underlying risk profiles of the borrowers. The observed distribution reflects a strategic approach by FinTech platforms to operate within a niche. This could also be a reflection of their focus on underserved segments who might typically require smaller loan amounts.

2.3 Geographical Distribution of Fintech

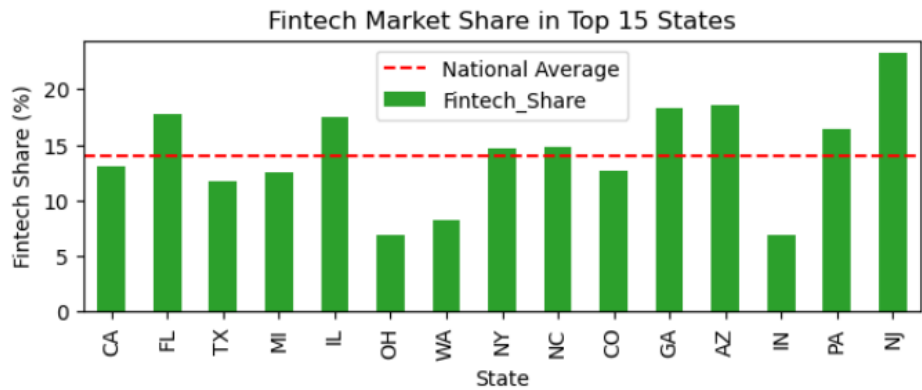


Figure 2.6: Fintech Market Share by State

The state based analysis of fintech adoption across 15 top states adds context to variations in FinTech market penetration across different U.S. states and even across regions. While the national average FinTech share is approximately 14.4%, states such as New Jersey, Arizona, and Georgia exhibit penetration rates considerably exceeding this benchmark, underscoring strong regional adoption of FinTech lending. Conversely, states like Wyoming and Mississippi, despite having smaller populations, also show high FinTech penetration, suggesting targeted FinTech strategies in underserved or digitally receptive markets. In contrast, states like Ohio, Washington State and Indiana demonstrate low FinTech engagement, pointing to regional disparities in the uptake of digital lending solutions. These variations are likely influenced by a confluence of factors, including a mix of urbanization, demographic structures (e.g., states like New Jersey, Florida and New York, which have high urban, minority, migrant, levels of educated and young demographics exhibit higher FinTech adoption rates, while states like Connecticut and Indiana which have larger elderly demographics tend to have lower fintech adoption), and the robustness of digital infrastructure, the success of FinTechs in certain states could be indicative of a more digitally literate population, proactive state regulatory environments, or a higher proportion of underserved communities receptive to alternative financial services.

3. Delinquency Performance

3.1 Delinquency Rate

Merging origination and servicing files allowed us longitudinal tracking of delinquency. We found that approximately 10.2% or 2873 of 28152 unique loans experienced some form of delinquency at some point within the availed time-frame.

Our finding further showed that 11.9% (476) of Fintech loans in the dataset turned delinquent compared to a slightly lower 9.93% (2397) among Non-fintech loans. This opposes the hypotheses from Fuster, A., Plosser, M., Schnabl, P., & Vickery, J. (2019). who suggested that fintechs' superior screening methods mitigated credit risk effectively.



3.2 Predictive Modeling - Random Forest Models

To evaluate predictive power of borrower metrics, we trained Random Forest classifiers using three single-variable models. While all the selected variables performed above the random classifier threshold, Fico score performs better on numerous indices.

Model I: FICO Score Only

- AUC: 0.73

The classification report shows some issues with the Random Forest model's performance: (confusion matrix attached below)

AUC Score (0.7315 ap 0.74) Good predictive power: Though the AUC shows the model can distinguish between classes better than random chance ($AUC > 0.5$), we believe this metric is misleading here. The high AUC is most likely a result of appropriate ranking of certain predictions rather than meaningful classification, as reflected by the failed precision/recall.

Delinquent: Precision = 0.00: this means that the model failed to correctly identify any actual delinquent cases (100% false positives for this class). Recall = 0.00: Model missed all actual delinquent cases (100% false negatives). F1-score = 0.00: Harmonic mean of precision/recall shows complete failure for Delinquent loans.

Non-Delinquent: Recall = 1.00: The model correctly predicted all non-delinquent cases, but this does not matter because of the extreme class imbalance (see Support). Precision = 0.00; all predictions of non-delinquent loans were in error (likely because there were no predicted cases of delinquent loans).

Confusion Matrix All Predictions as Non-Delinquent: The model predicted 284,312 cases as non-delinquent (true negatives) and 0 as delinquent, missing every one of the 2,575 actual delinquent cases. This shows the model defaults to the majority class, due to severe imbalance.

Model II: Debt-to-Income (DTI) Ratio Only

- AUC: 0.61

The model has moderate performance with significant limitations due to class imbalance (256K non-delinquent vs. 3.6K delinquent cases). Notable observations:

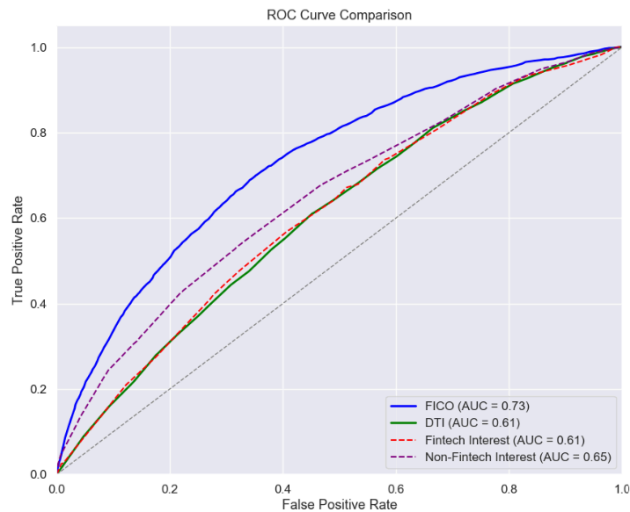
AUC (0.6077 ap 0.61) is slightly better than random (0.5), which indicates the model has weak ability to rank delinquent cases higher than non-delinquent cases. This agrees with the low recall for non-delinquent cases.

Class 0 (Non-Delinquent): Recall = 1.00 means that the model detects all non-delinquent cases (no false negatives). Precision = 0.70: 30% of predicted "non-delinquent" cases are delinquent (high false positives).

Class 1 (Delinquent): Precision = 0.40: Only 40% of predicted delinquent cases are correct (60% false positives). Recall = 0.40: Model misses 60% of actual delinquent cases (serious for risk management). F1-score = 0.40: Poor balance between precision and recall for minority class.

Macro vs. Weighted Averages: Macro avg ($F1 = 0.50$): Assigns equal weight to both classes, masking poor performance for Class 1.

Some Issues with the results: High False Negatives (FN): 2,145 delinquent cases misclassified as non-delinquent (serious for risk assessment). No FP for Class 0: Likely due to threshold bias towards the majority class.



Graph 3.1: ROC-CURVE.

FICO's superior predictive power aligns with credit risk theory, borrowers with lower scores have higher delinquency odds. The 13% AUC gap between FICO and DTI implies credit history matters more than current debt to income metric. Similar conclusions were reached by (Risbabh 2024; Fuster et al, 2019), who found that fintech lenders dynamically adjust pricing based on sophisticated borrower assessments, while non-fintech lenders often rely on coarser credit scoring methods.

3.3 Interest Rate Predictive Models (Stratified)

We further examine the predictive power of interest rates stratified by lender type:

- Fintech AUC (Interest Rate Model): 0.71
- Non-Fintech AUC (Interest Rate Model): 0.64

Higher AUC for fintechs could suggest that their interest rates more accurately reflect underlying risk, indicating more precise risk-based pricing.

4. Discussion and Conclusions

4.1 Comparison to Literature

Our findings align with some existing themes from recent empirical research, for instance; we found that fintech interest rates are slightly higher than non-fintech interest rates (Buchak, et al, 2018, Cornelli et al, 2020).

However, we also found some disparities, for instance, our analysis did not explicitly find that fintech's advanced technology would lead to lower rates of delinquencies (Buchak, et al, 2018; Cornelli et al, 2020; Fuster et al, 2018; Jagtiani et al, 2019)

However it could be implied from the comparative lower FICO scores and lower loan sizes of Fintech lenders, that fintechs providers service borrowers from excluded communities.

4.2 Conclusion

Risk-Based Pricing Accuracy:

FinTech:

- AUC of 0.71 for interest-rate models vs. 0.64 for traditional lenders.
- Interest rates more accurately predict delinquency across entire portfolios, reflecting better risk discrimination.

Classification Report:

Both had ideal precision/recall (1.00) on stratified test samples (n=2,000; 28 delinquent loans), however, under ideal conditions (artificially balanced classes), not representative of real-world data (10% true delinquency).

In essence we found that while interest rates correctly predict delinquency in balanced lender samples for both lenders, higher AUC for FinTechs means the model better captures true risk gradients in real-world imbalanced data.

Overall Lender Performance: In addition to loan performance, we use other metrics like financial inclusion, risk-based pricing, et cetera.

a) Loan Performance (Delinquency):

Traditional Lenders Win:

Lower raw delinquency (9.93% vs. 11.9%).

- Possibly an effect of lending to safer borrowers (looking at the avg. FICO score 748.51 vs. 733.72).

b) Financial Inclusion:

FinTechs Lead the Way:

- Twice more share of near-prime loans (8.9% vs. 4.7%).
- Focus on underserved groups (lower FICO, smaller loans: \$100K-\$200K).

c) Risk-Based Pricing:

FinTechs slightly edges non-fintech

- Slightly higher interest rates closer aligned with the true outcome (AUC difference: +0.07).
- Verifies their use of alternative data for proper risk estimation.

d) Operational Models:

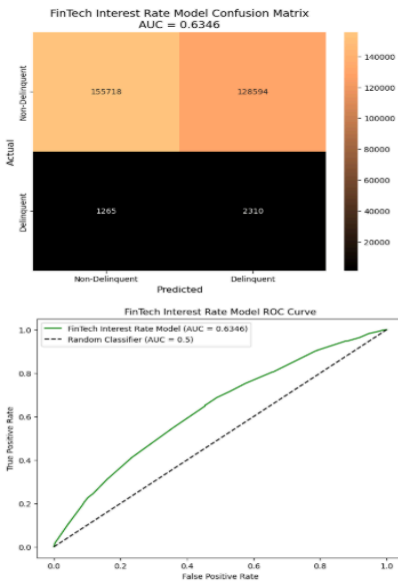
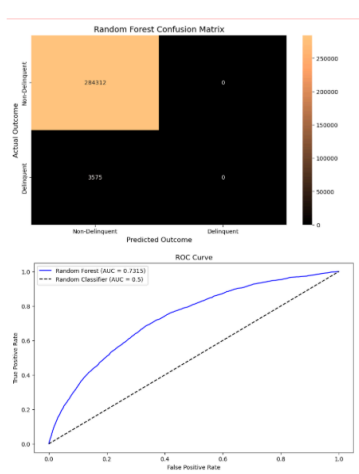
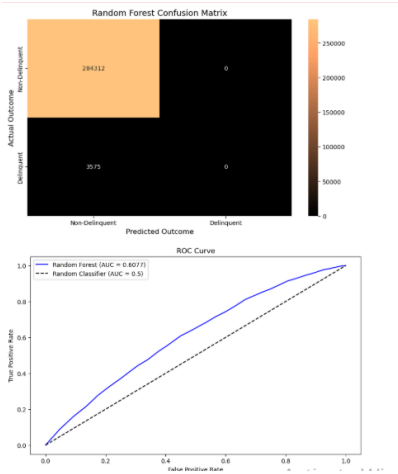
- FinTechs: probably utilise vertical integration (e.g., Quicken Loans servicing their own loans) which possibly leads to consistency and control.
- Traditional Lenders: Segmented third-party servicing probably leads to flexibility and specialization.

4.3 Limitations

- Detailed borrower-level income, employment, gender, immigration status, race, age and asset data were unavailable.
- Due to the long duration ending from 2017Q2 till 2020Q9, increased delinquencies could have been as a result of loss of livelihood as a result of covid-19 pandemic
- Not every alternative lending source would be documented in the Freddie Mac database, hence this might not be the most accurate representation of the mortgage market
- Essential and possibly revealing information like first-time borrower-flag, income and gender were unavailable, this meant that we were unable to ascertain if the premise that fintech lenders primarily provide financial services to underserved groups.
- The Random Forest classifier, exhibited high sample bias as the minority class data was not represented due to class imbalances, which impacted the generalizability of the model's predictions.

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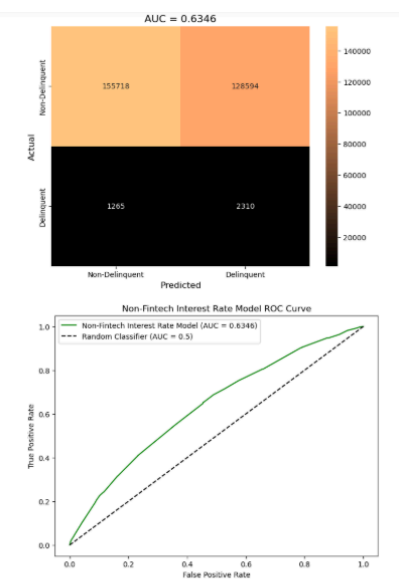


FinTech Interest Rate Model Classification Report:

	precision	recall	f1-score	support
0	0.99	0.55	0.71	284312
1	0.62	0.65	0.63	3576

accuracy 0.58 0.58 0.55 287887
macro avg 0.58 0.60 0.37 287887
weighted avg 0.98 0.55 0.70 287887

AUC Score: 0.6346



Non-FinTech Interest Rate Model Classification Report:

	precision	recall	f1-score	support
0	0.99	0.55	0.71	284312
1	0.62	0.65	0.63	3576

accuracy 0.58 0.58 0.55 287887
macro avg 0.58 0.60 0.37 287887
weighted avg 0.98 0.55 0.70 287887

AUC Score: 0.6346